THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

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ANSWER KEY & MARKING SCHEME SET 1

Chapter 10: **The Other Side of Zero** | Class 6 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** -7	The opposite has the same distance from 0 on the other side.
2	**(c)** -4	Add 1.
3	**(b)** -4	Subtract 1.
4	**(a)** 5	Use the number line or tokens.
5	**(b)** 2	The number to the right on the number line is greater.
6	**(c)** 6	Distance is never negative.
7	**(c)** negative integers	Definition
8	**(b)** 9	The opposite has the same distance from 0 on the other side.
9	**(d)** 1	Add 1.
10	**(c)** -1	Subtract 1.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Opposite of 5 is -5 (1); Opposite of -5 is 5 (1).	1+1
12	Successor $= -3$ (1); Predecessor $= -5$ (1).	1+1
13	Use the number line (1); $= -3$ (1).	1+1
14	Opposite of 11 is -11 (1); Opposite of -11 is 11 (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Subtracting an integer is the same as adding its opposite (1); $5 + (-8)$ (1); $= -3$ (1).	1+1+1
16	Negative numbers are smaller than 0 and 0 is smaller than positive numbers (1); Ascending: -7, -2, 0, 3, 5 (1); Further from 0 on the negative side means smaller (1).	1+1+1
17	New temperature $= -3 + 8$ (1); 5°C (1); State units (1).	1+1+1
	OR: Subtracting an integer is the same as adding its opposite (1); $-3 + (-4)$ (1); $= -7$ (1).	1+1+1
18	Negative numbers are smaller than 0 and 0 is smaller than positive numbers (1); Ascending: -9, -4, -1, 2, 6 (1); Further from 0 on the negative side means smaller (1).	1+1+1
	OR: New temperature $= -5 + 9$ (1); 4°C (1); State units (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	After going down: $2 - 5 = -3$ (1); After going up: $-3 + 1 = -2$ (1); Final floor -2 (1); Below ground (2).	1+1+1+2
	OR: Descending: 6, 3, -1, -4 (1); Smallest -4, largest 6 (1); Sum $= 2$ (1); Mark each at its distance from 0 (2).	1+1+1+2
20	After going down: $3 - 7 = -4$ (1); After going up: $-4 + 2 = -2$ (1); Final floor -2 (1); Below ground (2).	1+1+1+2
	OR: Descending: 5, 2, -3, -7 (1); Smallest -7, largest 5 (1); Sum $= -2$ (1); Mark each at its distance from 0 (2).	1+1+1+2

Mistakes to avoid:

- Saying $-7 > -3$ because 7 is greater than 3; on the number line -7 lies to the left of -3.
- Writing $-4 - (-6) = -10$; the correct value is $-4 + 6 = 2$.
- Dropping the sign of the bigger number when adding integers with different signs, e.g. $(-9) + 5 = 4$ instead of -4.
- Taking the successor of a negative integer as one smaller (successor of -1 is 0, not -2).
- Finding a fall or rise as initial − final in the wrong order; use final − initial and read the sign.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

ANSWER KEY & MARKING SCHEME SET 2

Chapter 10: **The Other Side of Zero** | Class 6 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** -9	Use the number line or tokens.
2	**(d)** -2	The number to the right on the number line is greater.
3	**(a)** 13	Distance is never negative.
4	**(b)** to the right	Definition
5	**(c)** -12	The opposite has the same distance from 0 on the other side.
6	**(a)** -11	Add 1.
7	**(c)** 7	Subtract 1.
8	**(c)** 2	Use the number line or tokens.
9	**(d)** 0	The number to the right on the number line is greater.
10	**(c)** 8	Distance is never negative.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Successor $= -8$ (1); Predecessor $= -10$ (1).	1+1
12	Use the number line (1); $= -9$ (1).	1+1
13	Opposite of 16 is -16 (1); Opposite of -16 is 16 (1).	1+1
14	Successor $= 1$ (1); Predecessor $= -1$ (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Subtracting an integer is the same as adding its opposite (1); $7 + 2$ (1); $= 9$ (1).	1+1+1
16	Negative numbers are smaller than 0 and 0 is smaller than positive numbers (1); Ascending: -8, -3, 0, 1, 8 (1); Further from 0 on the negative side means smaller (1).	1+1+1
17	New temperature $= -7 + 4$ (1); -3°C (1); State units (1).	1+1+1
	OR: Subtracting an integer is the same as adding its opposite (1); $-6 + 9$ (1); $= 3$ (1).	1+1+1
18	Negative numbers are smaller than 0 and 0 is smaller than positive numbers (1); Ascending: -6, -5, -2, 4, 7 (1); Further from 0 on the negative side means smaller (1).	1+1+1
	OR: New temperature $= -2 + 6$ (1); 4°C (1); State units (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	After going down: $1 - 4 = -3$ (1); After going up: $-3 + 6 = 3$ (1); Final floor 3 (1); Above ground (2).	1+1+1+2
	OR: Descending: 8, 1, -2, -6 (1); Smallest -6, largest 8 (1); Sum $= 2$ (1); Mark each at its distance from 0 (2).	1+1+1+2
20	After going down: $4 - 6 = -2$ (1); After going up: $-2 + 3 = 1$ (1); Final floor 1 (1); Above ground (2).	1+1+1+2
	OR: Descending: 7, 4, -5, -9 (1); Smallest -9, largest 7 (1); Sum $= -2$ (1); Mark each at its distance from 0 (2).	1+1+1+2

Mistakes to avoid:

- Saying $-7 > -3$ because 7 is greater than 3; on the number line -7 lies to the left of -3.
- Writing $-4 - (-6) = -10$; the correct value is $-4 + 6 = 2$.
- Dropping the sign of the bigger number when adding integers with different signs, e.g. $(-9) + 5 = 4$ instead of -4.
- Taking the successor of a negative integer as one smaller (successor of -1 is 0, not -2).
- Finding a fall or rise as initial − final in the wrong order; use final − initial and read the sign.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

ANSWER KEY & MARKING SCHEME SET 3

Chapter 10: **The Other Side of Zero** | Class 6 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** neither positive nor negative	Definition
2	**(c)** 15	The opposite has the same distance from 0 on the other side.
3	**(d)** 10	Add 1.
4	**(d)** -11	Subtract 1.
5	**(c)** 0	Use the number line or tokens.
6	**(a)** -9	The number to the right on the number line is greater.
7	**(c)** 20	Distance is never negative.
8	**(a)** -20	The opposite has the same distance from 0 on the other side.
9	**(b)** 0	Add 1.
10	**(a)** 14	Subtract 1.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Use the number line (1); $= -5$ (1).	1+1
12	Opposite of 24 is -24 (1); Opposite of -24 is 24 (1).	1+1
13	Successor $= -17$ (1); Predecessor $= -19$ (1).	1+1
14	Use the number line (1); $= 7$ (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Subtracting an integer is the same as adding its opposite (1); $0 + (-7)$ (1); $= -7$ (1).	1+1+1
16	Negative numbers are smaller than 0 and 0 is smaller than positive numbers (1); Ascending: -10, -5, 0, 5, 10 (1); Further from 0 on the negative side means smaller (1).	1+1+1
17	New temperature $= -9 + 12$ (1); 3°C (1); State units (1).	1+1+1
	OR: Subtracting an integer is the same as adding its opposite (1); $-12 + 5$ (1); $= -7$ (1).	1+1+1
18	Negative numbers are smaller than 0 and 0 is smaller than positive numbers (1); Ascending: -15, -8, -3, 9, 12 (1); Further from 0 on the negative side means smaller (1).	1+1+1
	OR: New temperature $= -4 + 10$ (1); 6°C (1); State units (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	After going down: $2 - 8 = -6$ (1); After going up: $-6 + 5 = -1$ (1); Final floor -1 (1); Below ground (2).	1+1+1+2
	OR: Descending: 9, 2, -3, -8 (1); Smallest -8, largest 9 (1); Sum $= 1$ (1); Mark each at its distance from 0 (2).	1+1+1+2
20	After going down: $5 - 9 = -4$ (1); After going up: $-4 + 2 = -2$ (1); Final floor -2 (1); Below ground (2).	1+1+1+2
	OR: Descending: 5, 3, -1, -6 (1); Smallest -6, largest 5 (1); Sum $= -1$ (1); Mark each at its distance from 0 (2).	1+1+1+2

Mistakes to avoid:

- Saying $-7 > -3$ because 7 is greater than 3; on the number line -7 lies to the left of -3.
- Writing $-4 - (-6) = -10$; the correct value is $-4 + 6 = 2$.
- Dropping the sign of the bigger number when adding integers with different signs, e.g. $(-9) + 5 = 4$ instead of -4.
- Taking the successor of a negative integer as one smaller (successor of -1 is 0, not -2).
- Finding a fall or rise as initial − final in the wrong order; use final − initial and read the sign.

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